

1. Write a Program to demonstrate Command Line Arguments.

```
1 public class commandline {
2     public static void main(String[] args) {
3         // Check if arguments are provided
4         if(args.length == 0) {
5             System.out.println("No arguments provided");
6         }
7         else{
8             System.out.println("Arguments received:");
9             for(int i = 0; i < args.length; i++) {
10                 System.out.println("Argument " + (i+1) + ": " + args[i]);
11             }
12         }
13     }
14 }
```

2. Write a Program to Create Student Information Using Array

```
1 class Student {
2     String name;
3     int age;
4
5     Student(String name, int age) {
6         this.name = name;
7         this.age = age;
8     }
9
10    void display(){
11        System.out.println("Name: " + name + ", Age: " + age);
12    }
13 }
14 public class StudentArray {
15     public static void main(String[] args) {
16         Student[] students = {
17             new Student("Alice", 20),
18             new Student("Bob", 22)
19         };
20         for (Student s : students) {
21             s.display();
22         }
23     }
24 }
```

3. Write a Program in Java to Implement User Defined Package

```
1 package mypackage;
2 public class myclass
3 {
4     public void display()
5     {
6         System.out.println("Hello , This is Demo of Package...");
7     }
8 }
9
10 //compilation of package in cmd
11 //javac -d . myclass.java
12 //-d denotes destination and . denotes current folder
```

```
1 import mypackage.myclass;
2 public class packagedemo
3 {
4     public static void main(String[] args)
5     {
6         myclass obj = new myclass();
7         obj.display();
8     }
9 }
```

4. Write a Program to Demonstrate Various String Functions

```
1 class stringfun{
2     public static void main(String args[]) {
3         String s1 = new String("Java programming");
4         String s2 = "JAVA programming";
5
6         System.out.println("Length: " + s1.length());
7         System.out.println("Character at 5: " + s1.charAt(5));
8         System.out.println("Concatenation: " + s1.concat(" Language"));
9         System.out.println("Equals: " + s1.equals(s2));
10        System.out.println("Upper Case: " + s1.toUpperCase());
11        System.out.println("Lower Case: " + s1.toLowerCase());
12        System.out.println("Substring: " + s1.substring(5, 16));
13        System.out.println("Replace: " + s1.replace('a', 'A'));
14    }
15 }
```

5. Write a Program in Java to implement Default and Parameterized Constructor

```
1  class Student
2  {
3      int rollNo;
4      String name;
5
6      Student()//Default
7      {
8          this.rollNo = 000;
9          this.name = "Unknown";
10     }
11     Student (int r,String n)//parameterized
12     {
13         this.rollNo = r;
14         this.name = n;
15     }
16     Student(Student s)//copy
17     {
18         this.rollNo = s.rollNo;
19         this.name = s.name;
20     }
21     void printst()
22     {
23         System.out.println(this.rollNo);
24         System.out.println(this.name);
25     }
26 }
27 public class classobjectcon {
28     public static void main(String[] args) {
29         Student s1 = new Student(1,"SHABBIR");
30         s1.printst();
31         Student s2 = new Student(2,"Taher");
32         s2.printst();
33         //Student s3 = new Student(s1.rollNo, s2.name);
34         Student s3 = new Student(s1);
35         s3.printst();
36     }
37 }
```

6. Write a Program in Java to Implement All type of Inheritance



```
1  //Single Inheritance
2  class Sname {
3      String name = "RAMAN";
4  }
5  class Student extends Sname
6  {
7      int age = 20;
8      void display(){
9          System.out.println("Name: "+name+", Age: "+ age);
10     }
11 }
12 public class SingleInheritance {
13     public static void main(String[] args) {
14         Student obj = new Student();
15         obj.display();
16     }
17 }
```



```
1  //Multi-Level Inheritance
2  class Sname {
3      String name = "RAMAN";
4  }
5  class Stid extends Sname
6  {
7      int age = 20;
8  }
9  class Student extends Stid
10 {
11     void display() {
12         System.out.println("Name: "+name+", Age: "+ age);
13     }
14 }
15 public class MultilevelInheritance {
16     public static void main(String[] args) {
17         Student obj = new Student();
18         obj.display();
19     }
20 }
```



```
1 //Hierarchical Inheritance
2 class Sname {
3     String name = "RAMAN";
4 }
5 class Student1 extends Sname
6 {
7     int age = 15;
8     void display1(){
9         System.out.println("Name: "+name+", Age: "+ age);
10    }
11 }
12 class Student2 extends Sname
13 {
14     int age = 20;
15     void display2(){
16         System.out.println("Name: "+name+", Age: "+ age);
17    }
18 }
19 public class HierarchicalInheritance {
20     public static void main(String[] args) {
21         Student1 obj1 = new Student1();
22         Student2 obj2 = new Student2();
23         obj1.display1();
24         obj2.display2();
25     }
26 }
```

7. Write a program in Java to Demonstrate Exception Handling



```
1 class exceptionhand{
2     public static void main(String args[]) {
3         try {
4             int a = 10, b = 0;
5             int c = a / b;
6             System.out.println("Result: " + c);
7         }
8         catch(ArithmeticException e) {
9             System.out.println("Exception Occurred: " + e);
10        }
11        finally {
12            System.out.println("Cannot perform Division");
13        }
14    }
15 }
```

8. Write a program in Java to Demonstrate Method Overriding & Overloading

```
1  class Parent {
2      void display() {
3          System.out.println("Display method of Parent class");
4      }
5  }
6  class Child extends Parent {
7      void display() {
8          System.out.println("Display method of Child class");
9      }
10     void data(int age)
11     {
12         System.out.println("Age = "+age);
13     }
14     void data(String name , int age)
15     {
16         System.out.println("Name = "+name+" Age = "+age);
17     }
18 }
19 public class overridingloading
20 {
21     public static void main(String args[]) {
22         Child obj = new Child();
23         obj.display();
24         obj.data(18);
25         obj.data("RAMAN",18);
26     }
27 }
```

9. Write a Program in Java to Demonstrate 2D Array

```
1  public class twodarray
2  {
3      public static void main(String[] args)
4      {
5          String[][] arr = {
6              {"Shabbir", "S145"},
7              {"Raman", "D845"}
8          };
9          for(int i=0;i<2;i++)
10         {
11             for(int j=0;j<2;j++)
12             {
13                 if(j==0)
14                     System.out.print("Name" + (i+1) + " = " + arr[i][j]);
15                 else
16                     System.out.println(" ,Registration ID" + (i+1) + " = " + arr[i][j]);
17             }
18             System.out.println();
19         }
20     }
21 }
```

10. Write a Program in Java to Demonstrate Use of Super Keyword



```
1  class Parent {
2      int num = 100;
3      Parent() {
4          System.out.println("Parent class constructor");
5      }
6      void display() {
7          System.out.println("This is the parent class method.");
8      }
9  }
10 class Child extends Parent {
11     int num = 200;
12     Child() {
13         super(); // Calls parent class constructor
14         System.out.println("Child class constructor");
15     }
16     void display() {
17         super.display(); // Calls parent class method
18         System.out.println("This is the child class method.");
19     }
20     void showNumbers() {
21         System.out.println("Child num: " + num);
22         System.out.println("Parent num: " + super.num);
23     }
24 }
25 public class superkey {
26     public static void main(String[] args) {
27         Child obj = new Child();
28         obj.display(); // Calls overridden method
29         obj.showNumbers(); // Shows variable values
30     }
31 }
```

11. Write a Program in Java to Demonstrate Scanner Class



```
1  import java.util.Scanner;
2
3  public class scannerdemo {
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6
7          System.out.println("Enter the Number : ");
8          int n = sc.nextInt();
9          if(n%2==0)
10         {
11             System.out.println("Number is Even");
12         }
13         else {
14             System.out.println("Number is Odd");
15         }
16     }
17 }
```

12. Write a Program in Java to Demonstrate Date and Time Functions



```
1  import java.util.Date;
2
3  class datetime{
4      public static void main(String args[]) {
5          Date d = new Date(); // Current date and time
6          System.out.println("Current Date & Time: " + d);
7      }
8  }
```


13. Write a Program in Java to Demonstrate Interfaces

```
1  interface Animal {
2      void eat();    // Abstract Method
3      void sleep(); // Abstract Method
4  }
5  class Dog implements Animal {
6      public void eat() {
7          System.out.println("Dog eats meat");
8      }
9
10     public void sleep() {
11         System.out.println("Dog sleeps at night");
12     }
13 }
14 public class interfacedemo
15 {
16     public static void main(String[] args)
17     {
18         Dog d = new Dog();
19         d.eat();
20         d.sleep();
21     }
22 }
```

14. Write a Program in Java to Demonstrate Abstract Class and Methods

```
1  abstract class Animal {
2      abstract void sound();
3      void eat() {
4          System.out.println("Animal eats food");
5      }
6      Animal()
7      {
8          System.out.println("This is Abstract Class Constructor");
9      }
10 }
11 class Dog extends Animal {
12     void sound() {
13         System.out.println("Dog barks");
14     }
15 }
16 public class abstractclass
17 {
18     public static void main(String[] args)
19     {
20         Dog obj = new Dog();
21         obj.eat();
22         obj.sound();
23     }
24 }
```

15. Write AWT Program in Java to create student's registration form and Demonstrate Different Events

```
import java.awt.*;
import java.awt.event.WindowAdapter;
import java.awt.event.WindowEvent;

public class Main extends Frame {
    public Main()
    {
        Label lbl1 = new Label("Enter Student's Full Name");
        lbl1.setBounds(22,30,153,20);
        add(lbl1);
        TextField txt1 = new TextField();
        txt1.setBounds(175,30,140,20);
        add(txt1);

        Label lbl2 = new Label("Gender: ");
        lbl2.setBounds(22,70,120,20);
        add(lbl2);
        CheckboxGroup genderGroup = new CheckboxGroup();
        Checkbox male = new Checkbox("Male", genderGroup, false);
        Checkbox female = new Checkbox("Female", genderGroup, false);
        male.setBounds(150,70,60,20);
        female.setBounds(225,70,60,20);
        add(male);
        add(female);

        Label lbl3 = new Label("Course : ");
        lbl3.setBounds(22,100,120,20);
        add(lbl3);
        Choice courseChoice = new Choice();
        courseChoice.add("BCA");
        courseChoice.add("BSc IT");
        courseChoice.add("MCA");
        courseChoice.add("B.Tech");
        courseChoice.setBounds(150,100,100,20);
        add(courseChoice);

        Label lbl4 = new Label("Address : ");
        lbl4.setBounds(22,140,120,20);
        add(lbl4);
        TextArea addressArea = new TextArea();
        addressArea.setBounds(150, 140, 200, 90);
        add(addressArea);
    }
}
```

```

Label lbl4 = new Label("Address : ");
lbl4.setBounds(22,140,120,20);
add(lbl4);
TextArea addressArea = new TextArea();
addressArea.setBounds(150, 140, 200, 90);
add(addressArea);

Button btn = new Button("Register");
btn.setBounds(22,250,53,26);
add(btn);

setSize(450,300);
setLayout(null);
setTitle("Student's Registration Form");
setVisible(true);

addWindowListener(new WindowAdapter() {
    @Override
    public void windowClosing(WindowEvent e) {
        dispose();
    }

    @Override
    public void windowActivated(WindowEvent e) {
        System.out.println("windowActivated");
    }

    @Override
    public void windowDeactivated(WindowEvent e) {
        System.out.println("windowDeactivated");
    }

    @Override
    public void windowClosed(WindowEvent e) {
        System.out.println("windowClosed");
    }
});
}

public static void main(String[] args) {
    Main obj = new Main();
}
}

```



```
1 Steps For AWT GUI in Java
2 public class Main extends Frame
3 Main Constructor
4 In Constructor
5 setSize
6 setLayout(null)
7 setTitle
8 setVisible
9 addWindowListener(new WindowAdapter)--->window closing---->dispose()
10
11 now all events must be before all this 4 functions given above
12 Button--->Button btn = new Button("name")--->btn.setBounds()-->add(btn)
13
14 Label--->Label lbl = new Label("text")--->lbl.setBounds()-->add(lbl)
15
16 TextField---->TextField txt TextField()---->txt.setBounds()---->add(txt)
17
18 Checkbox----->CheckboxGroup gender = new CheckboxGroup()
19             ----->Checkbox male = new Checkbox("Male", gender, false);
20             ----->male.setBounds()----->add(male)
21
22 Choice----->Choice courseChoice = new Choice();
23             ----->courseChoice.add("BCA");
24             ----->courseChoice.setBounds()----->add(courseChoice)
25
26 TextArea----->TextArea address = new TextArea()
27             ----->address.setBounds()----->add(address)
```

Student's Registration Form

Enter Student's Full Name

Gender: ☐ Male ☐ Female

Course :

Address :